RESULTS

PROCCESSING RESULTS USING FFT

1. FFT filters were used in place of Butterworth, but Median and Savgol filters are kept intact.

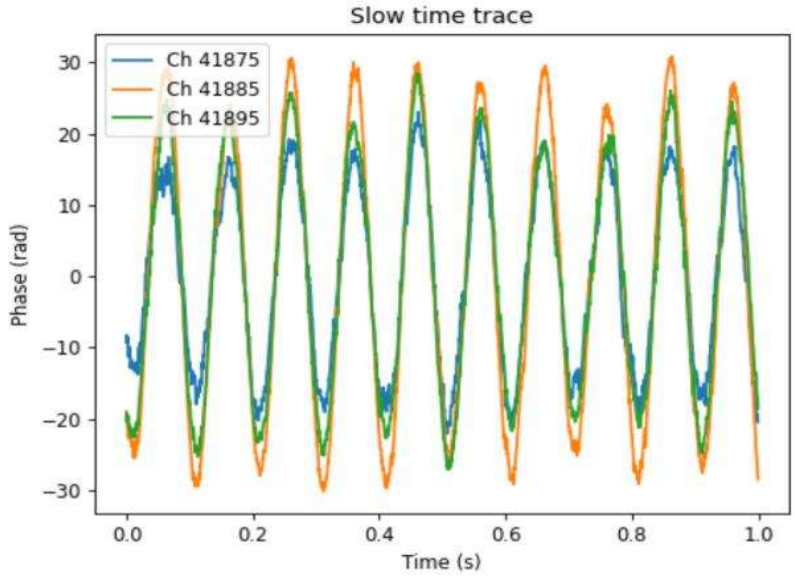


Figure 3 Slow time phase trace

This plot shows phase variation over time for selected channels (Ch 41875–41895). The oscillatory pattern indicates a periodic disturbance or event affecting the fiber at those spatial positions. The similarity across channels confirms that the perturbation is consistent and

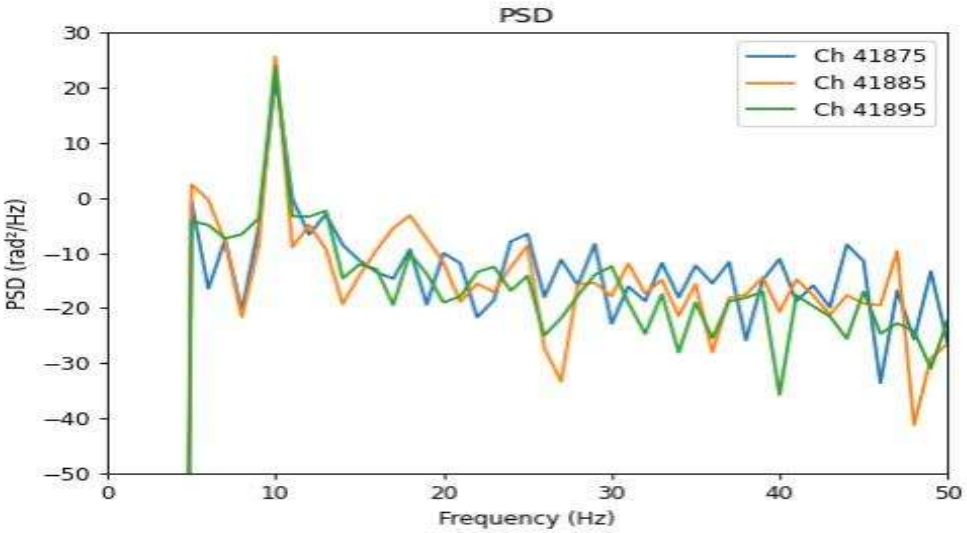


Figure 2 Power Spectral Density

localized across those distances. This is the primary time-domain signal used to analyze acoustic activity.

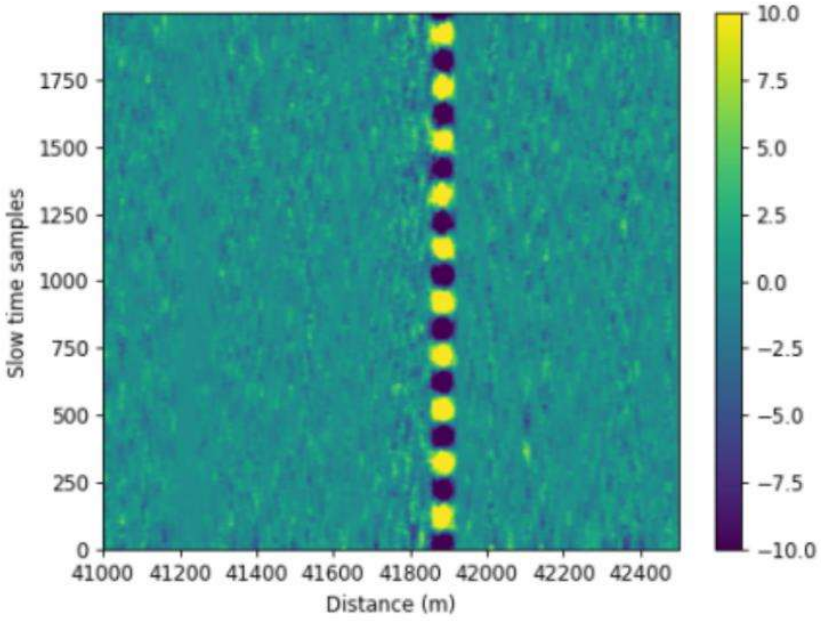


Figure 4 Waterfall Plot

A strong peak around 10 Hz is observed across all three channels, revealing the dominant frequency of the detected signal. The noise floor beyond 20 Hz is relatively uniform and low confirming that the signal of interest is narrowband and centered in the lower frequency range.

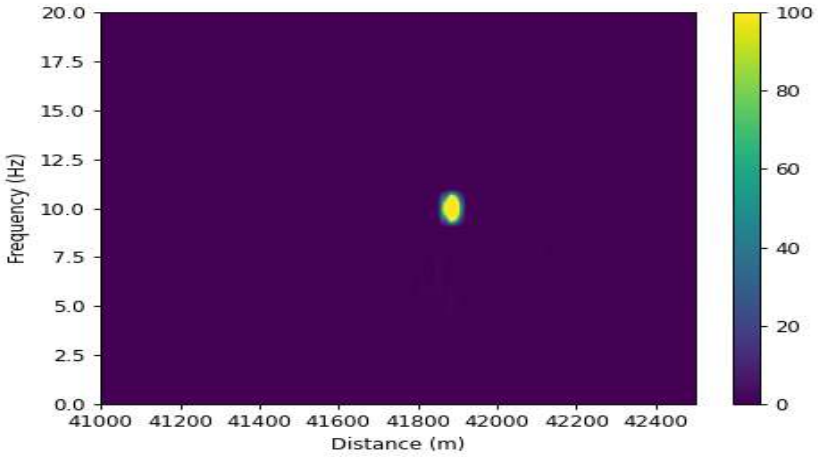


Figure 5 Spatiotemporal Phase Image

This shows power spectral density across distance and frequency. A strong, localized peak near 41867 m at ~10 Hz confirms the presence of a periodic event at that point on the fiber. The sharpness and isolation of the peak suggest high SNR and effective filtering.

SNR PLOT

To obtain the SNR plot, the data corresponding to this disturbed channel was extracted, and its power spectrum was computed. The signal power around the disturbance frequency (10 Hz) was compared to the average noise power across other frequency bins, forming a power vs frequency plot that quantified the signal-to-noise ratio at that specific location.

SNR for Butterworth = 28.82 dB

SNR for FFT = 28.57 dB

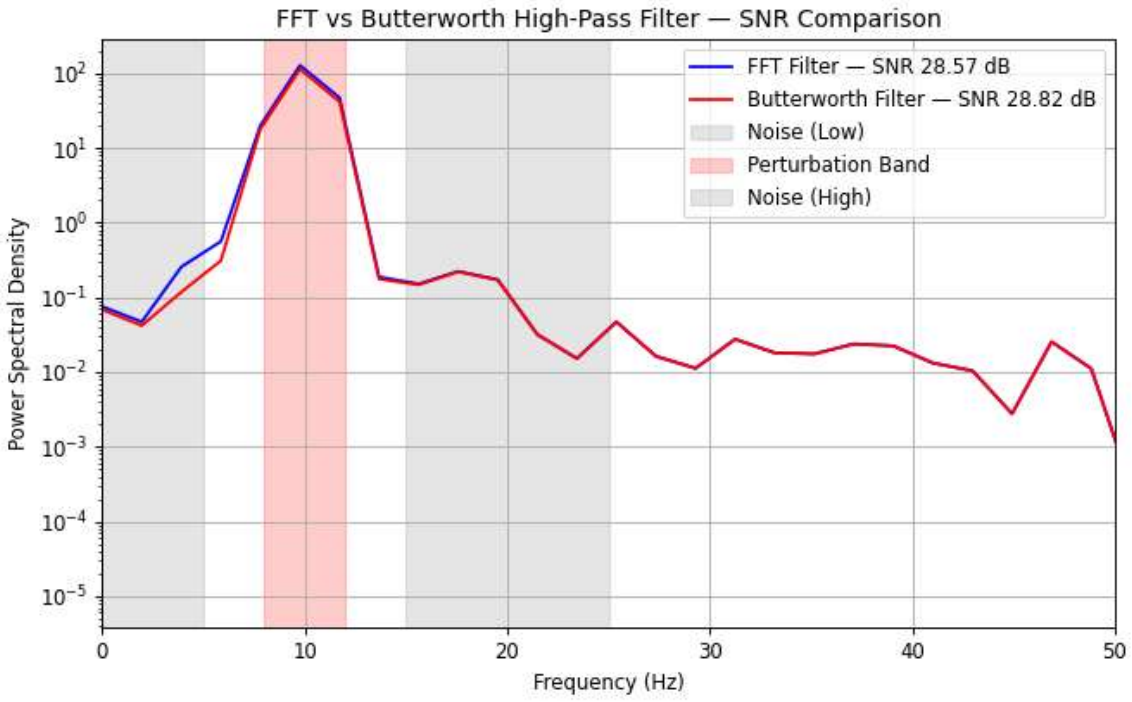


Figure 6 SNR plot

- **Power–Performance Trade-off –**

  The RTX 4000 focuses on maximizing throughput, whereas Jetson AGX Orin delivers balanced performance with significantly better energy efficiency, making it more suitable for embedded and real-time applications.

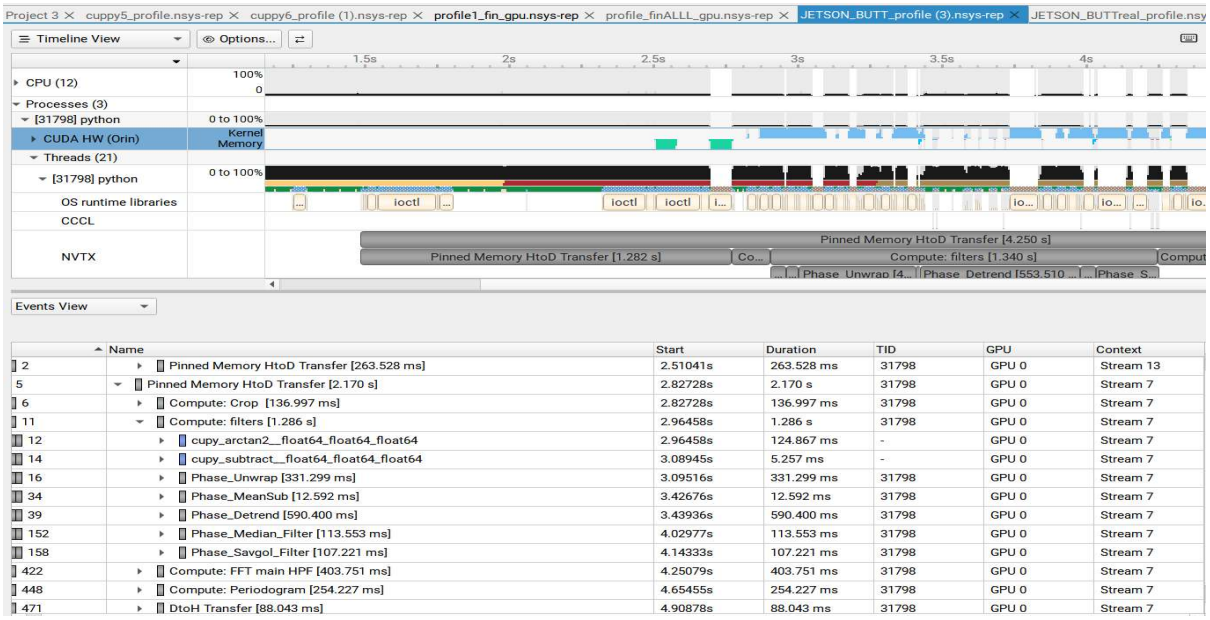


Figure 10 Jetson results using FFT

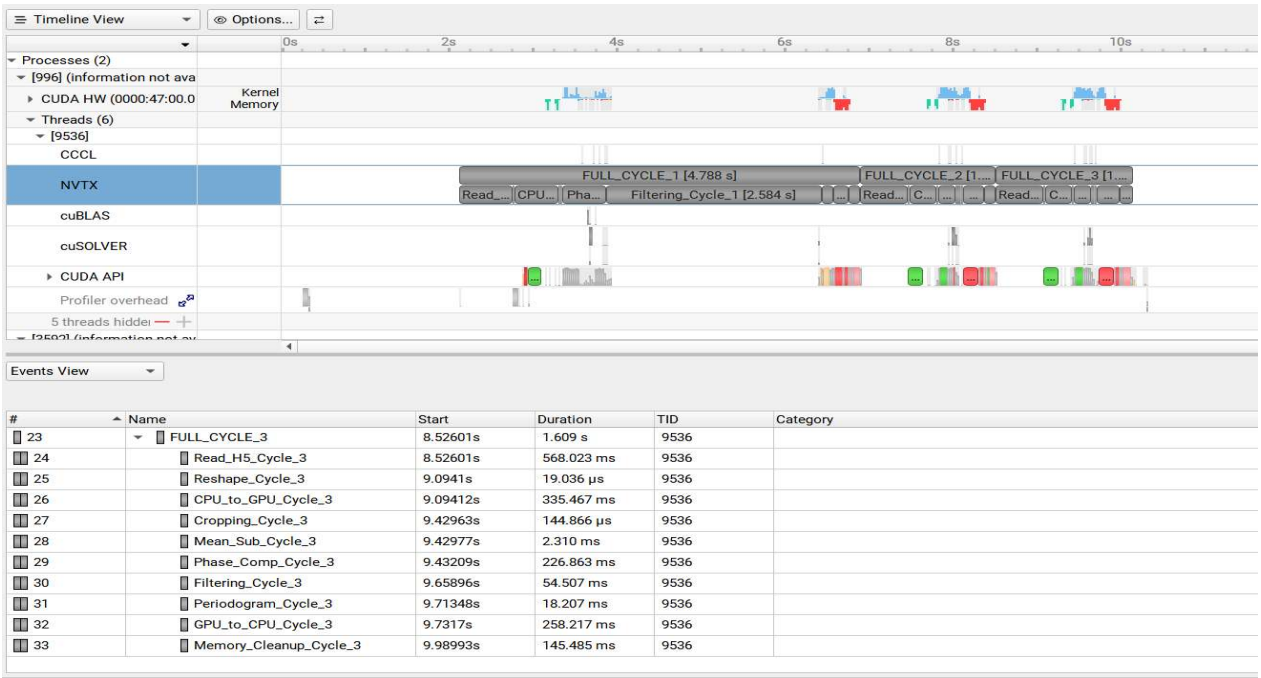


Figure 11 RTX results using FFT

**NOTE: Phase_Comp_Cycle contains median, savgol, and mean/arctan operations

RTX vs JETSON (Butterworth)

OPERATIONS	JETSON AGX ORIN (sec)	RTX Ada 4000(sec)
HtoD	0.24	0.32
Crop	0.12	88 x 10^-6
Median+Savgol+unwrap	1.24	0.22
Main Filter	5.95	0.053
Periodogram	0.17	0.017
DToH	0.09	0.26
TOTAL TIME	7.81	0.87

Table 2 RTX vs JETSON (Butterworth)

RTX vs JETSON (FFT filter)

OPERATIONS	JETSON AGX ORIN (sec)	RTX Ada 4000(sec)
HtoD	0.26	0.33
Crop	0.13	144 x 10^-6
Median+Savgol+unwrap	1.28	0.202
Main Filter	0.4	0.054
Periodogram	0.25	0.018
DToH	0.08	0.26
TOTAL TIME	2.32	0.864

Table 3 RTX vs JETSON (FFT filter)